

# Surya Bansal

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## EXPERIENCE

### JPMORGAN CHASE & CO. | Bangalore, IN

#### SOFTWARE ENGINEER II

June 24 – Present

- Improved the **Risk and Commodities** platform by delivering end-of-day risk and trading capabilities, helping front-office desks make timely decisions on positions exceeding **\$2B**.
- Designed a **scalable portfolio hierarchy** service using dynamic bucketing and parallel processing, **reducing processing latency by 60%** while improving downstream integration throughput for bullion and base metal traders.
- Built reliable **ETL and data integration pipelines** for credit-risk workflows across **5+ trading desks**, improving calculation accuracy and enabling automated downstream processing.
- Developed a React + Python reconciliation platform for **automated production validation**, reducing manual verification by **80%** and improving operational reliability.
- Built **agentic AI workflows** leveraging LLMs, prompt engineering and retrieval-based automation to resolve recurring engineering support requests, **reducing resolution time by 50%**.
- Developed **MCP servers** for Confluence exposing engineering utilities to AI agents for tool invocation and workflow automation.

### WALMART | Bangalore, IN

#### SOFTWARE ENGINEER

July 22 – June 24

- Developed a **PII Detection Service** from scratch using the Microsoft Presidio framework, enabling domain-based routing and delivering highly accurate, customized redaction results, saving the firm **\$5M annually**.
- Onboarded multiple tenants with Nginx reverse proxy servers and scaled the service through data preprocessing, achieving a **90% reduction in downtime** in the PII service.
- Developed a metrics dashboard using **Kafka and BigQuery** to monitor Cerbro Platform services, significantly enhancing real-time data analysis and improving operational efficiency.
- Engineered a redaction system automating legal document processing, delivering **\$2M in annual savings** with an **80% reduction in manual redaction** time.
- Owned the end-to-end development**, testing, deployment, and maintenance of the Soteria Redaction Platform. Evaluated cost-effective enterprise licenses and partnered with cross-functional teams to drive platform adoption across the organization.
- Built a high-throughput notification microservice for One Walmart serving over **100K daily impressions**, implementing caching, asynchronous processing and performance tuning for low-latency delivery..
- Built **GenAI prototypes** using prompt engineering, LLM APIs, and enterprise workflows for document summarization and intelligent automation, expanding AI adoption across the team.

### FAMPAY (YC S19) | Bangalore, IN

#### SOFTWARE ENGINEERING INTERN

Dec 21 – June 22

- Contributed to scaling FamPay's (YC S19, Sequoia-backed) digital payments infrastructure.
- Developed an automated reward distribution system for **5M+ users** that increased efficiency in customer engagement.
- Enhanced the overall test suite and built restful APIs using Redis cache, Celery and Django REST.

### RACCHABANDA | San Diego(Remote), CA

#### SOFTWARE ENGINEERING INTERN

Aug 21 - Dec 21

- Developed a web conferencing app using React, Redux, Node.js and optimised video latency.
- Implemented **virtual and blur backgrounds** with Tensorflow.js model(bodypix) using Web Workers.
- Added support for type ORM to SQLite APIs by migrating to Prisma.

## PROJECTS

### CROP RECOMMENDATION APP

Nov 24 | Grace Hopper Sustainability AI Hackathon (Winner)

- Built an AI-driven mobile app that combined weather, soil, and market data to help farmers make informed crop choices and improve yield outcomes.
- Stack: React Native, Generative AI models using Llama ,Mistral AI 7B Instruct model, OpenAI, FastAPI, Data Sources like Weather API, Soil API, Yield API for real-time data

### SEARCH AI TOOL

April 24 | Chalaang 2.0 Hackathon (Top 5)

- Engineered a context-aware search solution powered by RAG, LLMs, and vector stores, improving the accuracy of answers retrieved from the web.

### PR REVIEW TOOL

Oct 23 | Walmart X Google GenAI Hackathon (Winner)

- Developed a PR review bot to automate the code review process by analyzing pull requests. Used technologies like Prompt Engineering, Google Vertex AI, and GenAI models: textbison, codebison.

### SMART CITY MOBILE APPLICATION

Sept 20 | The Global Hack (Winner)

- Developed a React Native mobile application, enabling users to give direct feedback to the local government about their health, sanitation, and social distancing in the area during COVID-19.
- Awarded a \$300 MIT Solve micro-grant (2021) and €5,000 under hackathon's E-Governance track (2020).
- Represented India and built a smart city MVP which was mentioned by the Ministry of Electronics and Information Technology and The Economic Times.

### REACT WHITEBOARD APPLICATION

April 21 | Medium Publication

- Built a mobile-first responsive whiteboard using React Hooks, Redux, and Canvas API, and authored a detailed Medium tutorial with 40k+ interactions.

## TECHNICAL SKILLS

### LANGUAGES

Python • TypeScript • JavaScript • C/C++ • SQL

### FRAMEWORKS & LIBRARIES

React • React Native • Node.js • FastAPI • Django • Redux • GraphQL

### TOOLS & DEVOPS

Docker • Kubernetes • Azure • Linux • Git • Nginx • Kafka • BigQuery • Figma

### AI/ML

LangChain • Vector Stores • RAG • Google Vertex AI • Microsoft Presidio • Prompt Engineering • MCP • Claude Code

## EDUCATION

### GURU GOBIND SINGH

#### INDRAPRASTHA UNIVERSITY, DELHI

B.Tech in Information Technology

2022 | CGPA: 9.08